shouldn’t be a shock to learn that string/M-theory adds another candidate to the list of invisible things that may fill “empty” space. But let’s not get cavalier. For each of the previous possibilities, we understand its impact on physics and how we might establish that it truly exists. Indeed, for two of the three— dark energy and quantum fluctuations—we’ve seen that strong evidence supporting their existence has already been gathered; evidence for the Higgs field is being sought at current and future accelerators. So what is the corresponding situation for life within a three-brane? If the braneworld scenario is correct, why don’t we see the three-brane, and how would we establish that it exists?

The answer highlights how the physical implications of string/M-THEORY in the braneworld context differ radically from the earlier “brane-free” (or, as they’re sometimes affectionately called, no-braner) scenarios. Consider, as an important example, the motion of light—the motion of photons. In string theory, a photon, as you now know, is a particular string vibrational pattern. More specifically, mathematical studies have shown that in the braneworld scenario, only open string vibrations, not closed ones, produce photons, and this makes a big difference. Open string endpoints are constrained to move within the three-brane, but are otherwise completely free. This implies that photons (open strings executing the photon mode of vibration) would travel without any constraint or obstruction throughout our three-brane. And that would make the brane appear *completely transparent—completely invisible*—thus preventing us from seeing that we are immersed within it.

Of equal importance, because open string endpoints cannot leave a brane, they are unable to move into the extra dimensions. Just as the wire constrains its beads and the pinball machine constrains its balls, our sticky three-brane would permit photons to move *only* within our three spatial dimensions. Since photons are the messenger particles for electromagnetism, this implies that the electromagnetic force—light—would be trapped within our three dimensions, as illustrated (in two dimensions so we can draw it) in Figure 13.3.

That’s an intense realization with important consequences. Earlier, we required the extra dimensions of string/M-theory to be tightly curled up. The reason, clearly, is that we don’t see the extra dimensions and so they must be hidden away. And one way to hide them is to make them smaller